Table 3. Thirteen BBJ discoveries and their FinnGen GD comparison

BBJ discovery gene	Chr	IVs, BBJ/FinnGen GD	BBJ OR (95% CI)	BBJ *P*	FinnGen GD OR (95% CI)	FinnGen GD *P*	Focused support	FinnGen GD PP.H4
HLA-A	6 (MHC)	4/4	1.98 (1.73–2.27)	2.60×10^{-23}	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	6 (MHC)	1/1	2.39 (1.95–2.93)	8.30×10^{-17}	1.90 (1.64–2.20)	1.41×10^{-17}	Yes	<0.001
CTLA4	2	1/2	0.18 (0.11–0.27)	5.45×10^{-15}	0.17 (0.08–0.36)	2.15×10^{-6}	Yes	0.254
TSHR	14	1/1	0.12 (0.07–0.21)	1.09×10^{-14}	0.08 (0.05–0.12)	3.19×10^{-32}	Yes	0.264
C4A	6 (MHC)	1/3	0.45 (0.35–0.57)	2.23×10^{-10}	0.41 (0.26–0.63)	4.97×10^{-5}	Yes	<0.001
HSD3B7	16 (16p11.2 cluster)	1/2	0.29 (0.18–0.47)	2.04×10^{-7}	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	6 (MHC)	2/3	1.49 (1.28–1.74)	3.14×10^{-7}	3.28 (2.09–5.17)	2.68×10^{-7}	Yes	<0.001
VKORC1	16 (16p11.2 cluster)	1/0	0.13 (0.06–0.29)	6.35×10^{-7}	NA	NA	NA	0.011
TNFSF14	19	1/1	0.63 (0.52–0.76)	1.48×10^{-6}	0.77 (0.65–0.92)	0.00361	Yes	0.303
PRSS36	16 (16p11.2 cluster)	1/1	6.00 (2.87–12.57)	2.02×10^{-6}	1.40 (1.00–1.96)	0.0490	No	0.063
MAPKAPK5	12	4/4	0.977 (0.967–0.987)	5.28×10^{-6}	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	6 (MHC)	1/1	1.25 (1.13–1.38)	6.86×10^{-6}	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	6	1/1	2.10 (1.51–2.91)	9.41×10^{-6}	1.62 (1.18–2.24)	0.00330	Yes	0.313

Genes passed the BBJ screen threshold ($P < 0.05/2{,}544$) and are listed by BBJ *P* value. Focused support in FinnGen GD requires the same direction and $P < 0.05/13$; 7 of 13 qualify. IVs are counted separately for each dataset because instrument availability differs. The five MHC genes and the three chromosome 16p11.2 genes are not independent signals; the latter are compatible with one regional association. *MAPKAPK5* is shown to three decimals to preserve its small effect. NA, not estimable (no retained instrument), not a negative result; *VKORC1* colocalization does not require an MR instrument. PP.H4, shared-variant posterior probability under $p12 = 10^{-5}$; bold, ≥0.80. Chr, chromosome of the strongest instrument; MHC, major histocompatibility complex; CI, confidence interval; IV, instrumental variable; MR, Mendelian randomization; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.